

Functions as values: iterators and streams

Block 2 — Thunks and streams: when is a value computed?

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Block 2: the three parts

In a strict language a value is computed where it is written. In this block we put a **function** in its place.

- | | | |
|---|---------------------------------------|---|
| 1 | Thunks
when? | We choose when the value is computed: when we call the function. |
| 2 | Sequences
how far? | We put the function inside a data structure. We choose how far a sequence is computed, and it can be without end. |
| 3 | Memoisation
how many times? | We keep the result of the function. We choose that the value is computed at most once. |

Thunks

How does a strict language wait?

Session 4: if written as a function

```
let my_if c t e = if c then t else e
```

OCAML

```
let rec fact n =  
  my_if (n = 0) 1 (n * fact (n - 1))
```

`fact 5` never returns. OCaml computes the two branches before it calls `my_if`.

```
let my_if c t e = if c then t () else e ()
```

OCAML

```
let rec fact n =  
  my_if (n = 0) (fun () -> 1)  
    (fun () -> n * fact (n - 1))
```

`fact 5` is 120. `my_if` calls one of the two functions.

A function that holds a computation and waits for its call is a **thunk**. In OCaml we write it `fun () -> e`.

A thunk, called twice

```
const t = () => {  
  console.log("computing");  
  return 6 * 7;  
};
```

```
console.log("the thunk is made");  
const a = t();  
const b = t();  
console.log(a, b);
```

JAVASCRIPT

Run

Reset

```
the thunk is made  
computing  
computing  
42 42
```

What does the program print, and in which order?

- When the thunk is made, its body is not run.
- At every call the body is run again.

The same program in OCaml

```
let t = fun () ->  
  print_endline "computing";  
  6 * 7
```

OCaml ↗

```
let () = print_endline "the thunk is made"  
let a = t ()  
let b = t ()
```

A thunk computes **at every call**. It is session 4's **call-by-name**, written by the program.

A left fold and a right fold: who holds the rest of the list?

```
let rec fold_left f acc xs =  
  match xs with  
  | [] -> acc  
  | x :: xs -> fold_left f (f acc x) xs
```

OCaml

Our function receives the accumulator and **one element**. The rest of the list, `xs`, stays with the fold.

```
let rec fold_right f xs z =  
  match xs with  
  | [] -> z  
  | x :: xs -> f x (fold_right f xs z)
```

OCaml

Our function receives one element and **the fold of the rest**, as its second argument.

Only with a **right fold** is our function given the rest, and can decide what becomes of it.

`fold_right` keeps one frame waiting per element; `fold_left` keeps none: its recursive call is a tail call.

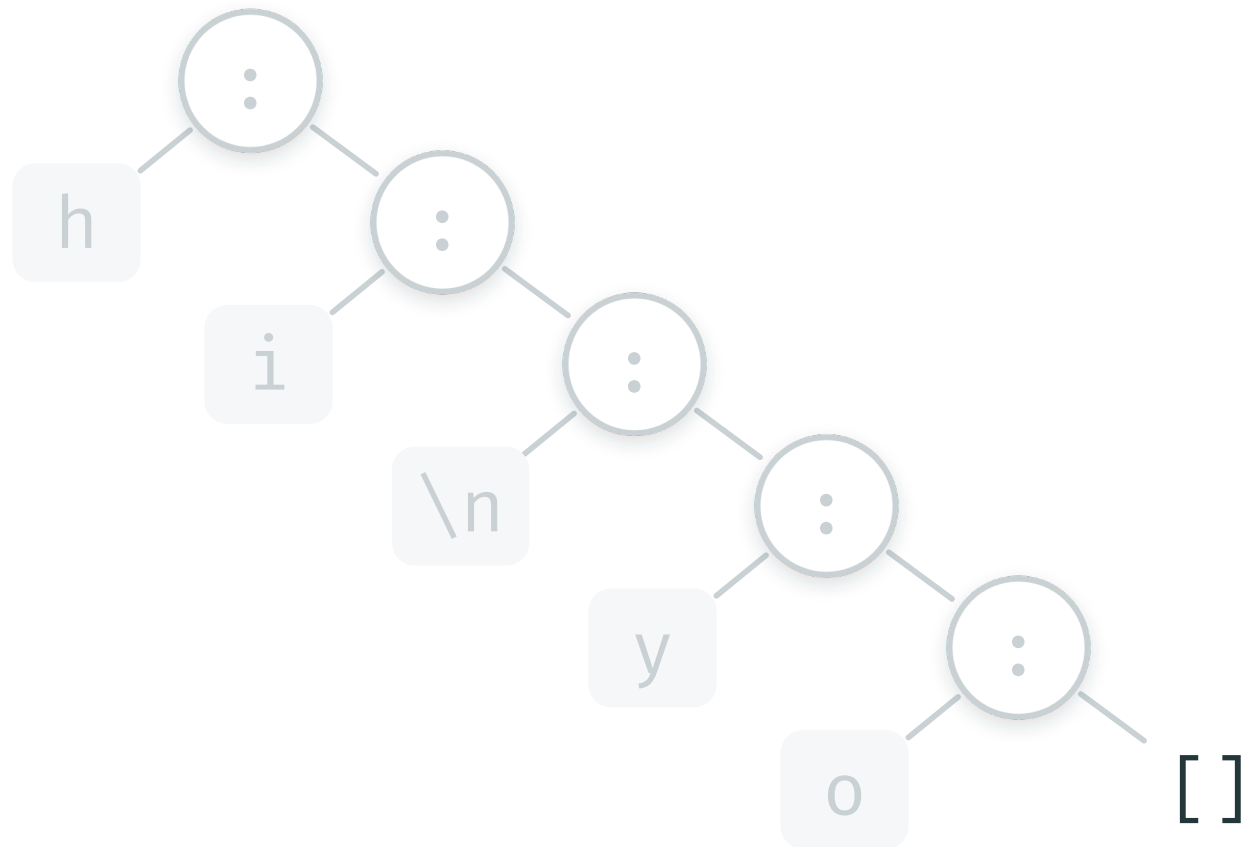
Block 1's exists, with the left fold

```
let exists p l = fold_left (fun found x -> found || p x) false l
```

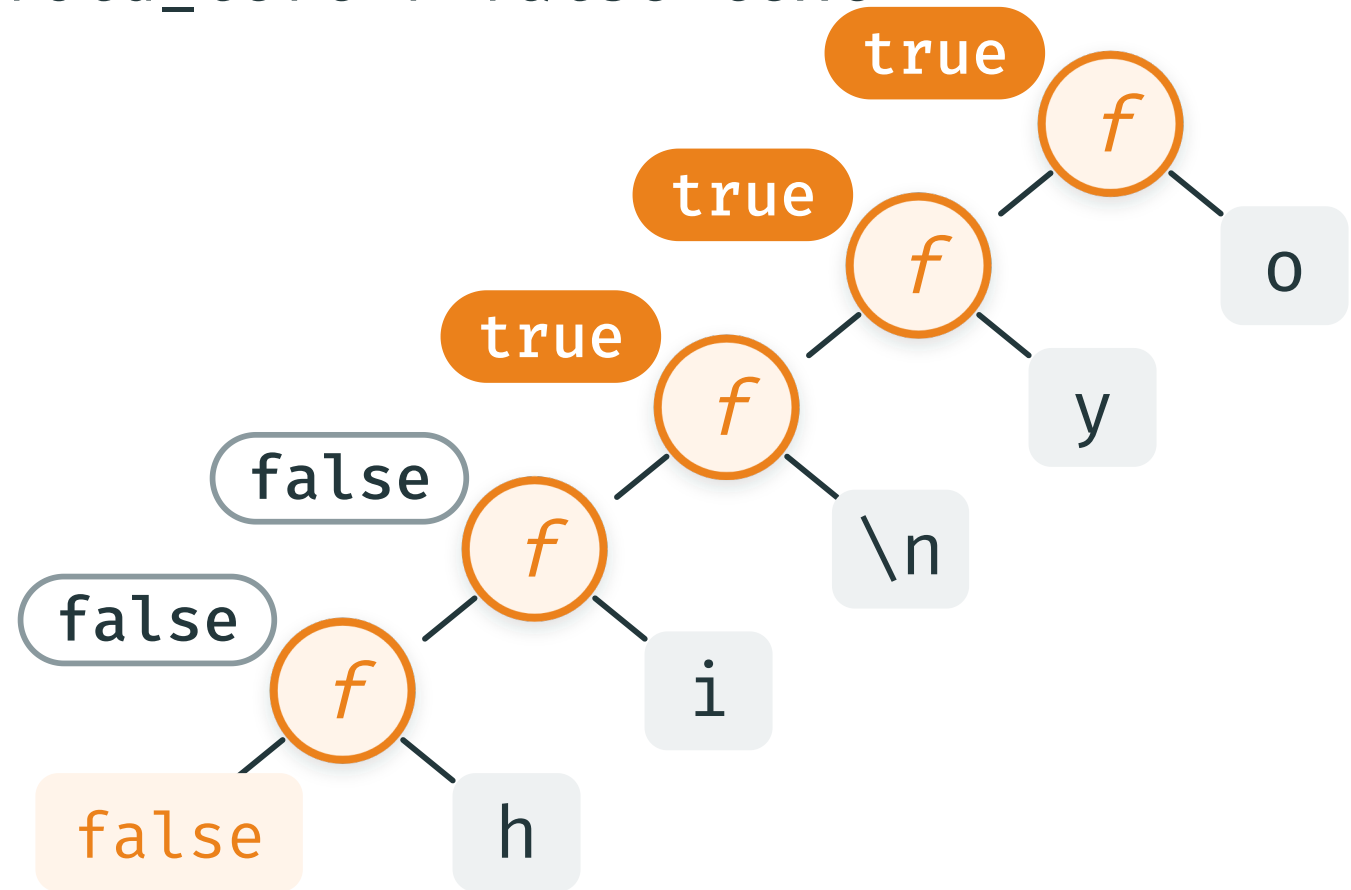
[OCaml ↗](#)

```
exists (fun c -> c = '\n') text
```

the list



fold_left f false text



Step Run Reset

f called 5 times

f was given one element at a time. The rest of the list never reached it: 5 calls

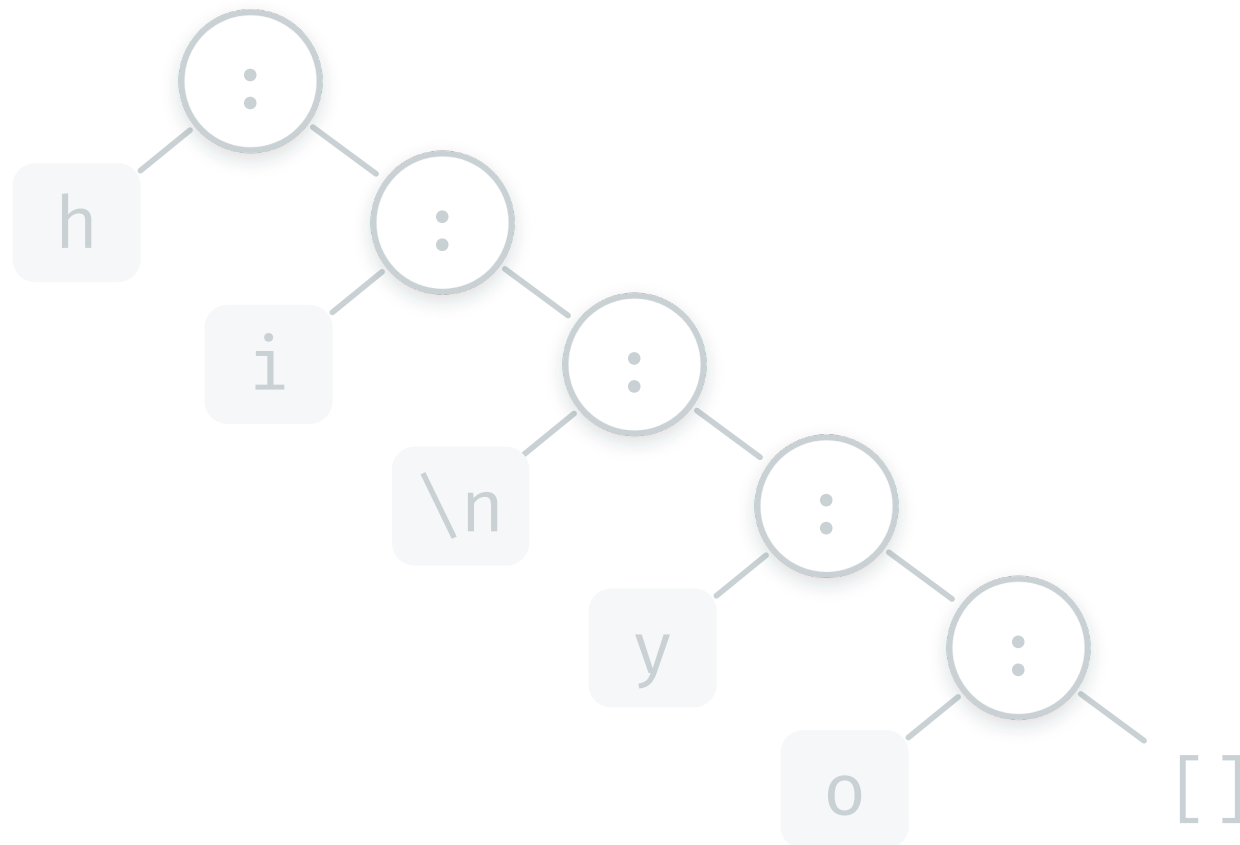
exists, with the right fold

```
let exists p l = fold_right (fun x rest -> p x || rest) l false
```

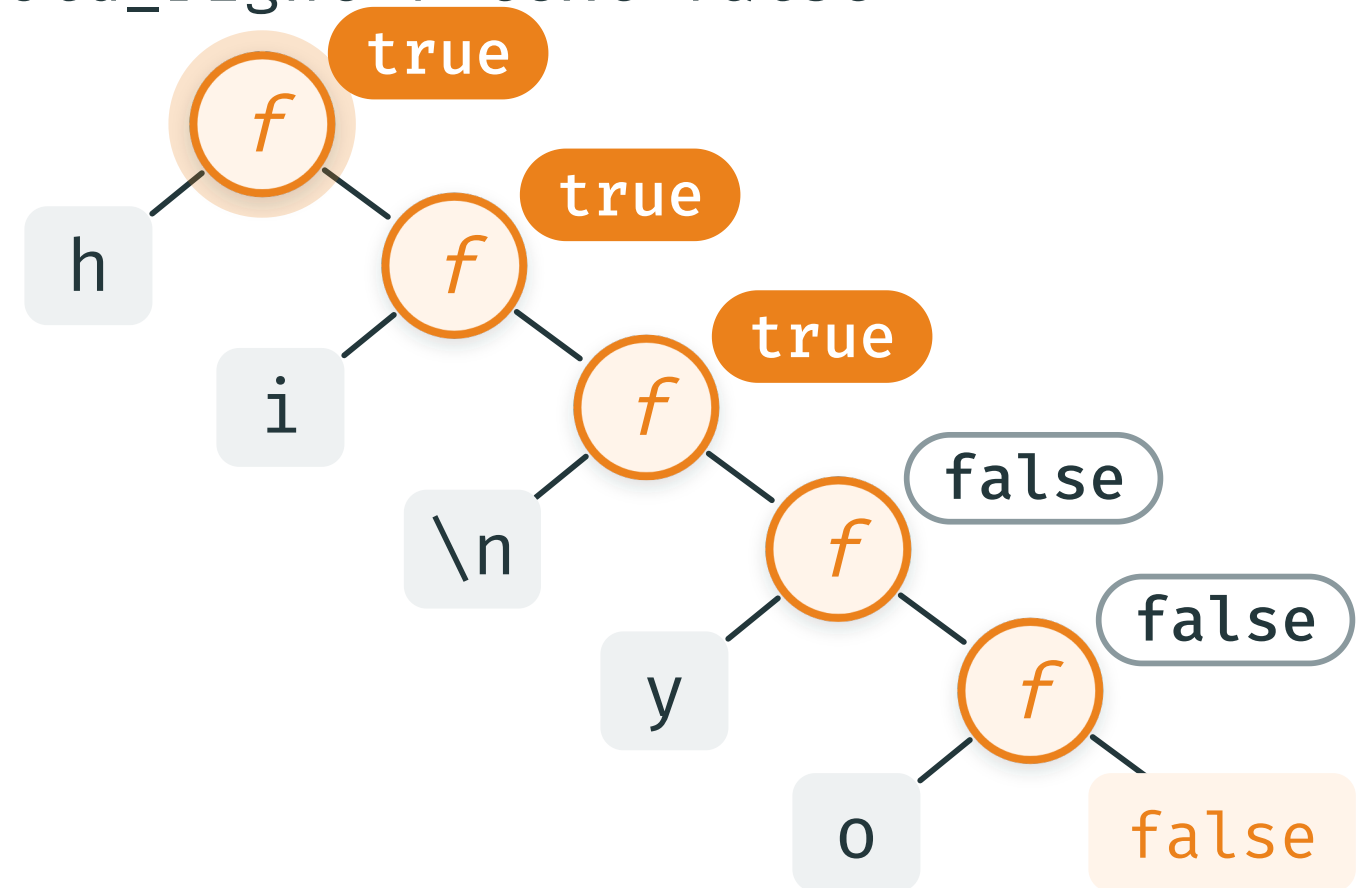
[OCaml ↗](#)

```
exists (fun c -> c = '\n') text
```

the list



fold_right f text false



Step Run Reset

f called 5 times

f is called on 'h'. Its second argument is true, already computed: it answers true

The right fold, with the rest as a thunk

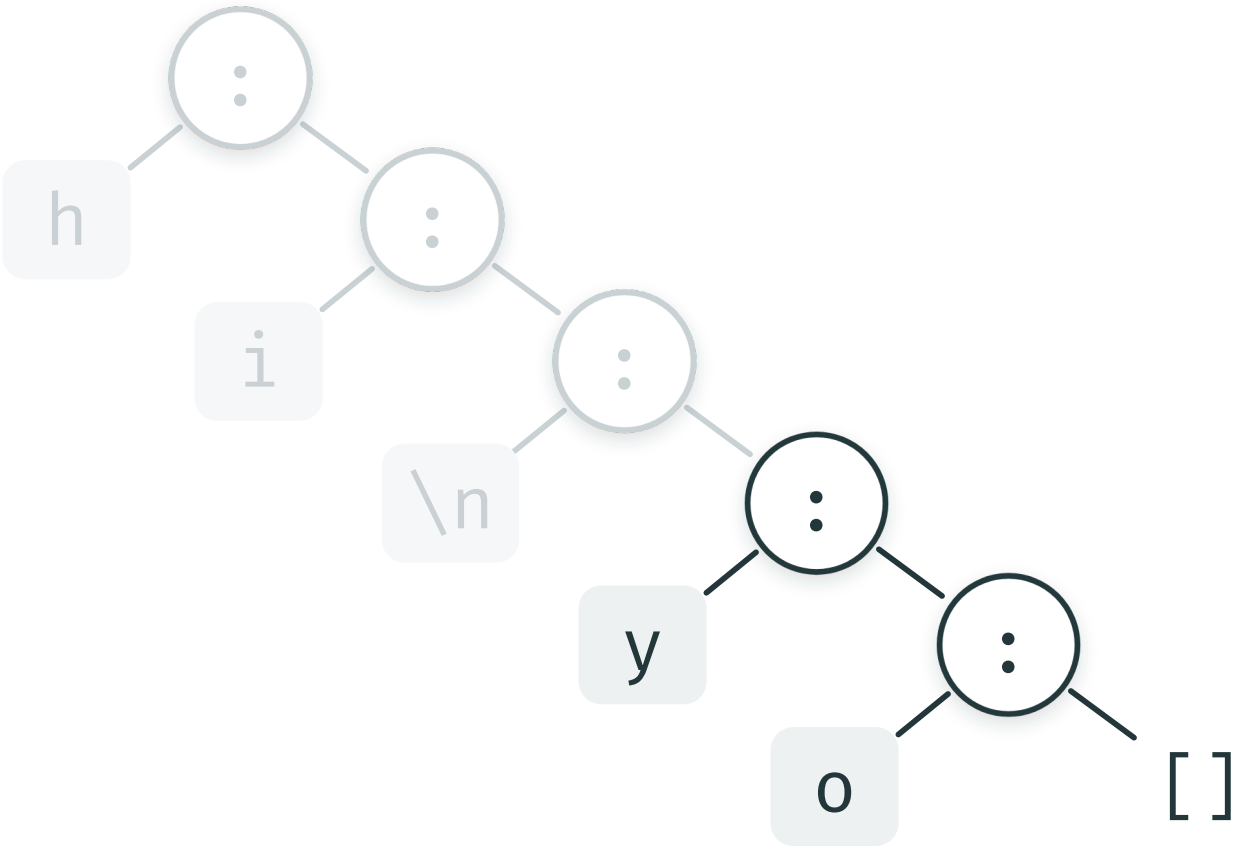
```
let rec fold_right f xs z = match xs with
| [] -> z
| x :: xs ->
    f x (fun () -> fold_right f xs z)
```

OCaml

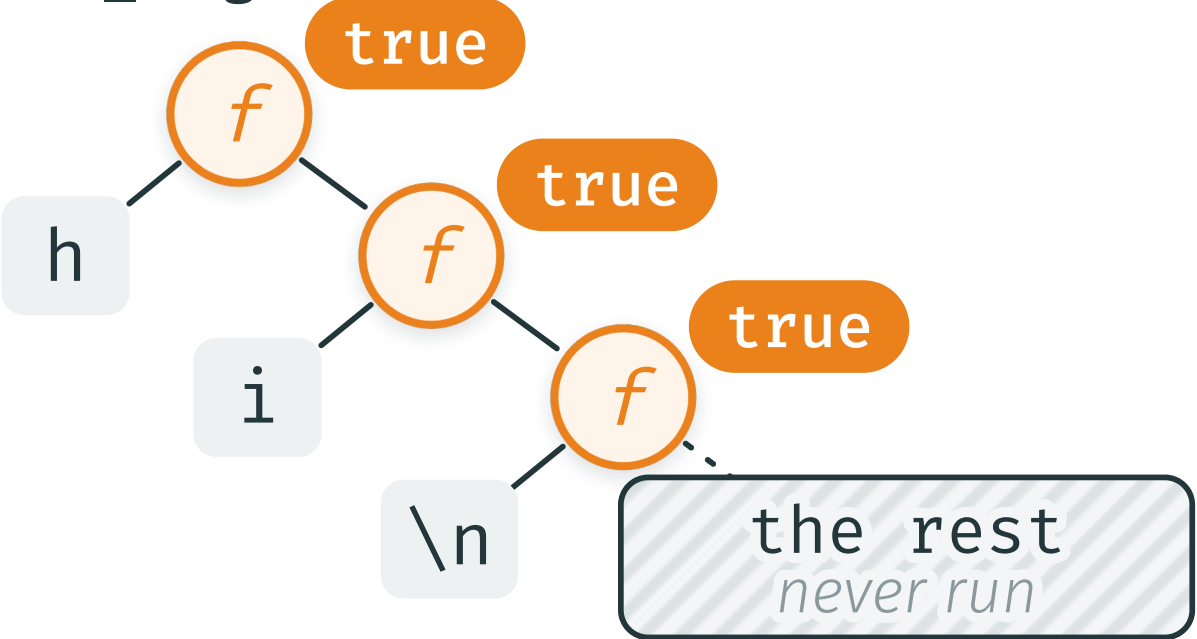
```
let exists p l =
  fold_right
    (fun x rest -> p x || rest ())
    l false
```

OCaml ↗

the list



fold_right f text false



The list of **all** the natural numbers

```
let rec from n = n :: from (n + 1)
```

OCaml ↗

```
let rec take n l =  
  if n = 0 then []  
  else match l with  
    | [] -> []  
    | x :: rest -> x :: take (n - 1) rest
```

```
take 5 (from 0)
```

Stack overflow during evaluation

```
from n = n : from (n + 1)
```

Haskell ↗

```
take n l =  
  if n == 0 then []  
  else case l of  
    [] -> []  
    x : rest -> x : take (n - 1) rest
```

```
take 5 (from 0)
```

[0,1,2,3,4]

The two programs are the same, line for line. Why does Haskell give an answer, and OCaml none?

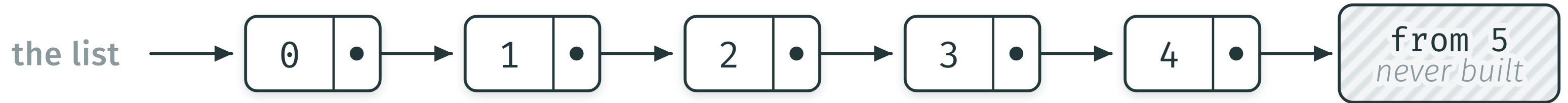
A list built **lazily**: when it is read

```
take 5 (from 0)
```

```
-- Haskell also writes from 0 as [0 ..]
```

HASKELL

take 0



Step Run Reset

cells built 5 cells in the list **no end**

take 0 reads nothing: the program ends. The rest of the list, from 5, is never built

Haskell builds a cell of the list when a function **reads** it, and not before. `take 5` reads five cells, so five cells are built.

The list of all the prime numbers

```
primes = sieve [2 ..]  
  where sieve (p : xs) = p : sieve [x | x <- xs, x `mod` p /= 0]
```

[Haskell ↗](#)

in Haskell

we read

as we know it

[2 ..]

the list 2, 3, 4, ... without end

from 2

p : xs

a list: its first element `p`, its rest `xs`

OCaml: `p :: xs`

[x | x <- xs, c]

the list of the `x` of `xs` for which `c` holds

Python: `[x for x in xs if c]`

x `mod` p /= 0

`x` is not a multiple of `p`

OCaml: `x mod p <> 0`

where

a definition written after its use

OCaml: `let ... in`, before

`sieve` keeps `p`, the first element, and calls itself on the rest **without the multiples of** `p`. The list in brackets is block 1's `filter`.

The sieve, on its first three lists

The first number of the list is prime. We keep it, we take its multiples out of the rest, and we do the same with what is left.

p = 2	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	...
p = 3		3		5		7		9		11		13		15		17		19		21		23		25	...
p = 5				5		7				11		13				17		19				23		25	...

```
take 10 primes = [2,3,5,7,11,13,17,19,23,29]
```

The sieve is applied to a list without end, and it calls itself on a list without end. Why does the program end?

Lazy evaluation, in Haskell

In Haskell no argument is computed before the call. It is computed when the function uses it, and at most once: session 4's **call-by-need**.

in OCaml, we write

in Haskell

Block 1	another fold, to stop before the end of a list	the right fold stops by itself
Block 2	a thunk for each value that must wait	every value waits, and is computed once
Session 6	two functions, for a search that can fail	the list of all the solutions, of which we read the first

In session 4 the language chose when an argument is computed. In session 5 **the program chooses**, with a function.

The two folds, on a list without end

```
foldr (\x rest -> x > 2 || rest)
      False [0 ..]
```

Haskell ↗

True

Our function is given the rest, not computed. At 3 it answers, and the cells after 3 are never built.

```
foldl (\found x -> found || x > 2)
      False [0 ..]
```

HASKELL

the program does not end

`foldl` gives the rest of the list to itself. It answers at the end of the list, and the list has no end.

Laziness delays what a function is given. In Haskell as in OCaml, only with the **right fold** is our function given the rest.

Sequences

How do we postpone the next element?

From a list to a sequence

```
type 'a list =  
  | []  
  | (::) of 'a * 'a list
```

OCaml

The rest of a list is a list: a **value**. It is computed before we read the first element.

```
type 'a node =  
  | Nil  
  | Cons of 'a * 'a seq
```

OCaml

```
and 'a seq = unit -> 'a node
```

The rest of a sequence is a **thunk**. It is computed when we call it.

A **sequence** (also called a stream)* is a thunk that gives one element and the rest of the sequence.

* CS3110, section 9.4. OCaml's library has this type as `Seq.t`.

nats and take

```
let rec nats n : int seq =  
  fun () -> Cons (n, nats (n + 1))
```

OCaml ↗

```
let rec take n (s : 'a seq) =  
  if n = 0 then []  
  else match s () with  
    | Nil -> []  
    | Cons (x, rest) ->  
      x :: take (n - 1) rest
```

`nats 0` is the sequence of all the natural numbers.

What does `take 5 (nats 0)` compute, and how many elements of `nats 0`?

`[0; 1; 2; 3; 4]` `produced=5`

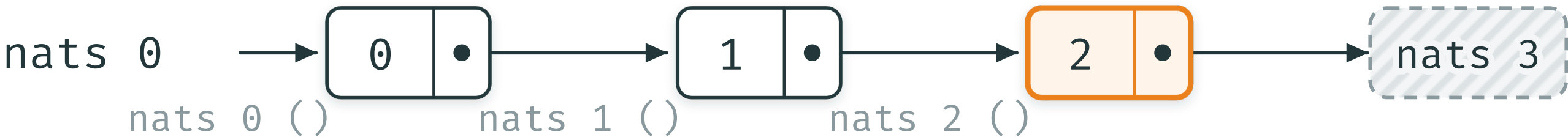
A sequence without end is a **value**: we can name it, pass it and keep it. Five of its elements are computed, because we ask for five.

A list and a sequence, drawn

a list: every cell is built before we read the first one



a sequence: a cell is built when we call the thunk



Step Run Reset

cells built 3

it builds one cell: the number 2, and the thunk nats 3

After three calls, three cells are built. The rest, nats 3, is a thunk that nobody has called.

map and filter on a sequence

```
let rec map f (s : 'a seq) : 'b seq =  
  fun () -> match s () with  
    | Nil -> Nil  
    | Cons (x, rest) ->  
      Cons (f x, map f rest)
```

OCaml ↗

`map f s` returns at once: its body is under
`fun () -> .`

In a `Cons`, the recursive call returns at once: it makes a function that **nobody has called yet**. The rest is computed when the consumer asks for it.

```
let rec filter p (s : 'a seq) : 'a seq =  
  fun () -> match s () with  
    | Nil -> Nil  
    | Cons (x, rest) ->  
      if p x  
      then Cons (x, filter p rest)  
      else filter p rest ()
```

OCaml

When `x` does not pass, `filter` asks for the next element itself.

How far is a sequence computed?

```
let even n = n mod 2 = 0
let square n = n * n

let squares =
  map square (filter even (nats 0))
```

OCaml ↗

```
produced=0
```

```
take 5 squares
```

OCAML

```
[0; 4; 16; 36; 64]    produced=9
```

How many elements has `nats 0` produced when `squares` is defined? And after `take 5`?

Nine numbers, from 0 to 8. Five of them are even.

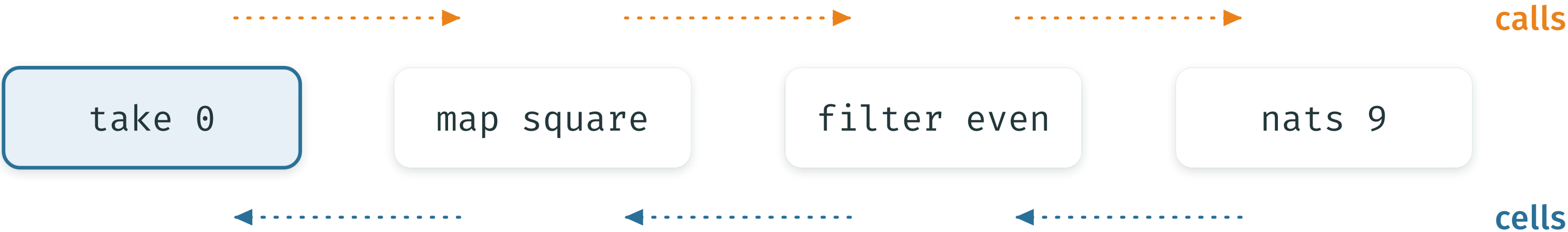
Defining `squares` computes no element. The consumer decides **how far** it is computed.

take 5 squares, step by step

```
= [0; 4; 16; 36; 64]
```

the consumer

the producer



the list built

- 0
- 4
- 16
- 36
- 64

Step Run Reset

cells built by nats 9

take 0 is the empty list: take calls nothing more, and nats 9 is never called

Stopping, by not asking

```
let rec exists p (s : 'a seq) =  
  match s () with  
  | Nil -> false  
  | Cons (x, rest) ->  
    p x || exists p rest
```

OCaml ↗

The text of block 1, `hi\nyo`, as a sequence of characters:

```
exists (fun c -> c = '\n') text
```

OCAML

```
found=true asked=3
```

With a sequence **we pull**. To stop, we do not ask for the next element.

The library: Seq

```
Seq.ints 0
|> Seq.filter (fun n -> n mod 2 = 0)
|> Seq.map (fun n -> n * n)
|> Seq.take 5
|> List.of_seq
```

[OCaml ↗](#)

```
[0; 4; 16; 36; 64]
```

`Seq.t` is the type that we wrote, and `Seq.ints 0` is our `nats 0`.

`x |> f` is `f x`. With it the line reads in the order the elements travel, from `ints` to `take`, and not from the inside out.

Memoisation

Can a value be computed at most once?

A thunk and a lazy value, each asked twice

```
let ticks = ref 0
```

OCaml ↗

```
let t = fun () -> incr ticks; 6 * 7
```

```
let a = t ()
```

```
let b = t ()
```

a=42 b=42 ticks=2

```
let ticks = ref 0
```

OCAML

```
let l = lazy (incr ticks; 6 * 7)
```

```
let a = Lazy.force l
```

```
let b = Lazy.force l
```

a=42 b=42 ticks=1

What is `!ticks` at the end of each program?

A thunk **delays**. A lazy value delays and **remembers**: it is session 4's **call-by-need**.

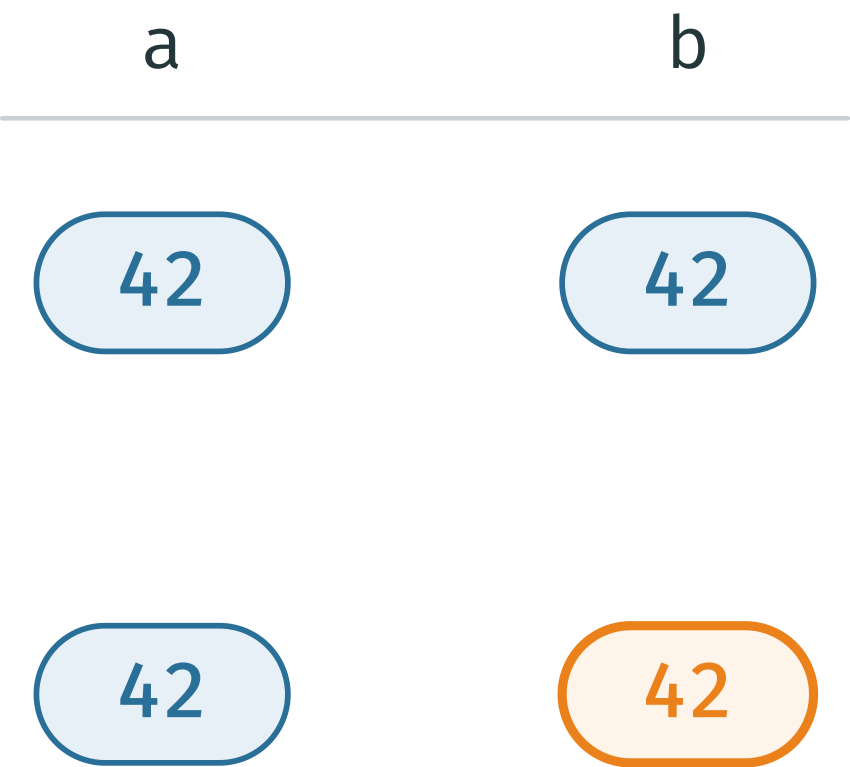
Okasaki 1998, chapter 4: a *stream* is part 2's sequence with a lazy value, not a thunk, for the rest.

A thunk and a lazy value, drawn

a thunk



a lazy value



Step Run Reset

b = Lazy.force l: 42 is read. Nothing runs

the body of the thunk has run 2 of the lazy value 1

lazy, made by hand

```
type 'a state =  
  | Delayed of (unit -> 'a)  
  | Value of 'a  
  
let delay f = ref (Delayed f)  
  
let force s = match !s with  
  | Value v -> v  
  | Delayed f ->  
    let v = f () in  
    s := Value v;  
    v
```

OCaml ↗

A lazy value is a reference that holds a thunk, and after the first **force** the value.

```
let s = delay (fun () ->  
  incr ticks; 6 * 7)  
  
let a = force s  
let b = force s
```

OCAML

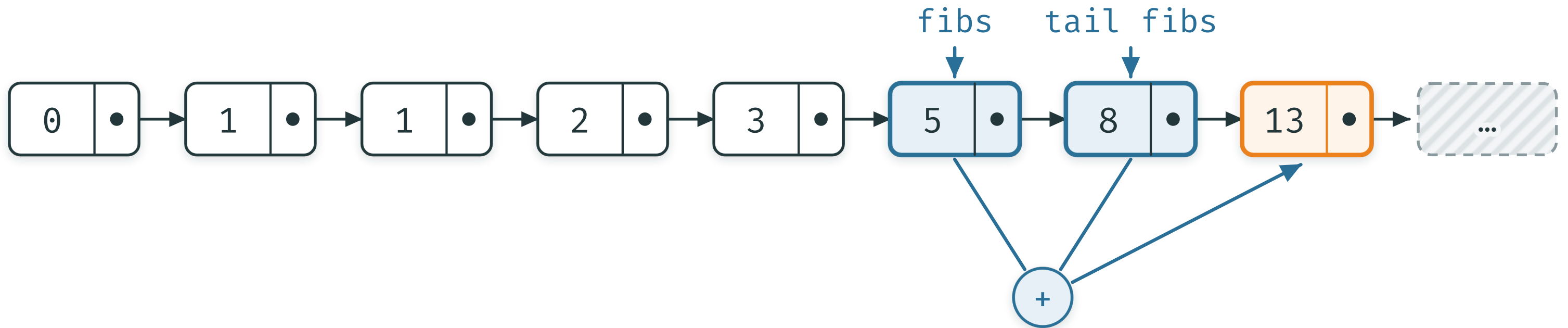
```
a=42 b=42 ticks=1
```

With **delay** we write the **fun () ->** ourselves. **lazy e** is a **keyword**: a function would receive **e** already computed.

With a table in the place of the reference, we keep one result for each argument. This is the **memoisation** of a function: see the appendix.

The Fibonacci numbers, as a list without end

```
fibs = 0 : 1 : zipWith (+) fibs (tail fibs)
```

[Haskell ↗](#)

[Step](#) [Run](#) [Reset](#)

additions, every cell kept 6 every cell computed again 46

cell 7 is asked for. The two cells it reads are already there: one addition, $5 + 8 = 13$

The list is read at two places, one cell apart, and their sum is its next cell. Each cell is computed **once**, and read twice.

fibs: how many additions?

```
type 'a node = Nil | Cons of 'a * 'a seq
and 'a seq = 'a node Lazy.t
(* zip_with and tail read their sequences with Lazy.force *)
let rec fibs = lazy (Cons (0, lazy (Cons (1, zip_with ( + ) fibs (tail fibs)))))
```

[OCaml ↗](#)

The same list in OCaml. To compute its first 25 numbers, how many additions are needed?

the rest of the sequence is

additions

a thunk, in OCaml

196 392

a `lazy` value, in OCaml

23

a list, in Haskell

23

With `lazy`, `fibs` and `tail fibs` **share** the numbers. In Haskell every value is lazy, by default.

Lab 2's queue, without memoisation

```
let pop = function
  | x :: front, rear ->
    Some (x, (front, rear))
  | [], rear ->
    match rev rear with
    | [] -> None
    | x :: front ->
      Some (x, (front, []))
```

OCaml ↗

Lab 2's queue, as we wrote it there: a front and a rear. When the front is empty, `pop` reverses the rear.

```
(* q: eight elements pushed *)
let a = pop q
let b = pop q
```

OCAML

```
a=1 b=1 reversals=2
```

The queue is persistent: `q` can be used twice. With this `pop`, the reversal is then paid twice.

With `lazy` the reversal is suspended inside `q`: the two pops share it, and it is **paid once**. We write that queue as homework, at the end of the warm-up's sheet.

Source: Okasaki 1998, chapter 6, the banker's queue.

Conclusion of block 2

	the question	the program decides, in OCaml	the language decides, in Haskell
1	When is a value computed?	When we call the thunk that holds it.	When it is used.
2	How far is a sequence computed?	As far as we ask: a sequence .	Every list is a sequence.
3	How many times?	Once, if we keep the result: <code>lazy</code> .	Once: every value is lazy, by default.

In this block a function held **a computation not yet run**: `unit -> 'a`. We call it to get a value.

In session 6 a function holds **the rest of the computation**: `'a -> 'r`. We call it to give a value.

Session 6: a promise is a value that is not there yet, together with what to do when it arrives.

Memoising a function

Can a function remember what it has computed?

fib 25, without memoisation

```
let rec fib n =  
  if n < 2 then n  
  else fib (n - 1) + fib (n - 2)
```

```
let a = fib 25  
let b = fib 25
```

OCAML

```
a=75025 b=75025 the body ran 485570 times
```

The second call computes everything again: a function keeps nothing from one call to the next. With `lazy (fib 25)` we keep one result, for the argument 25. Every other argument needs a lazy value of its own.

A lazy value keeps the result of **one computation**. To memoise a function is to keep a result **for each argument**: a table in the place of the reference.

memo: a function that keeps a table

```
let memo f =  
  let table = Hashtbl.create 16 in  
  fun x ->  
    match Hashtbl.find_opt table x with  
    | Some y -> y  
    | None ->  
      let y = f x in  
      Hashtbl.add table x y;  
      y
```

OCaml ↗

`memo f` makes a table, once, and returns a function.

- Called on `x`, the function looks `x` up in the table.
- If the result is there, it returns it, and `f` is not called.
- If not, it calls `f`, keeps the result, and returns it.

The function returned is a **closure with a state**, as block 1's counter. `lazy` is the same with one entry: the table of a function of no argument.

memo fib: the calls that the table sees

```
let rec fib n =  
  if n < 2 then n  
  else fib (n - 1) + fib (n - 2)  
  
let mfib = memo fib
```

[OCaml ↗](#)

	result	the body of <code>fib</code> ran
<code>mfib 25</code>	75025	242 785 times
<code>mfib 25</code> , again	75025	0 times

The second call is answered from the table. The first one is as slow as `fib 25`.

Inside `fib`, the two recursive calls are calls of `fib`, not of `mfib`. The table sees only the calls that **we** make.

memo_rec: the function receives the function to call

```
let fib' self n =  
  if n < 2 then n  
  else self (n - 1) + self (n - 2)
```

OCaml

`fib'` does not call itself: it has no `rec`. It calls `self`, which it receives.

```
memo_rec fib' 25 = 75025  
the body ran 26 times
```

```
let memo_rec f =  
  let table = Hashtbl.create 16 in  
  let rec g x =  
    match Hashtbl.find_opt table x with  
    | Some y -> y  
    | None ->  
      let y = f g x in  
      Hashtbl.add table x y;  
      y  
  in  
  g
```

OCaml ↗

`memo_rec` hands `g`, the function with the table, to `f` as its `self`. Every call goes **through the table**.

Functions of order 1, 2 and 3

	its type	order	
<code>fib</code>	<code>int -> int</code>	1	its argument is not a function
<code>memo</code>	<code>('a -> 'b) -> ('a -> 'b)</code>	2	its argument is a function
<code>fib'</code>	<code>(int -> int) -> (int -> int)</code>	2	
<code>memo_rec</code>	<code>(('a -> 'b) -> ('a -> 'b)) -> ('a -> 'b)</code>	3	its argument is a function whose argument is a function

The three ways to keep a result are one construction: a reference for **one value** (`lazy`), a list for **each position** (`fibs`), a table for **each argument** (`memo`).

Two exercises

1. Given `memo`, write `delay` and `force` of the slide “`lazy`”, made by hand” with it, and count the ticks.
2. Given `memo_rec` and the numbers of Pascal's triangle,

```
let rec choose (n, k) =  
  if k = 0 || k = n then 1  
  else choose (n - 1, k - 1) + choose (n - 1, k)
```

OCaml

write `choose'`, which receives the function to call, and count how many times each body runs for `(24, 12)`.

	result	the body ran
<code>choose (24, 12)</code>	2 704 156	5 408 311 times
<code>memo_rec choose' (24, 12)</code>	2 704 156	168 times